THE COMPOSITION PAIRING BETWEEN EXT AND THE DUAL EQUALS THE CUP PRODUCT UNDER THE IDENTIFICATION OF HOM WITH SECTIONS OF A TENSOR PRODUCT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement contains a commutative diagram typeset with Xy-pic; the retained chapter preamble supplies the native Xy-pic setup, so it compiles exactly as in the source.

Editorial note. The counted span cites nine cross-chapter labels (including an example and an equation), each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- Comparison of the two cup products. Stacks Project, Cohomology of Sheaves, Lemma lemma-second-cup-equals-first (tag 0A8J). Identifies the cup product used in the statement with the tensor product of the representing maps.
- Duality for perfect complexes: H^0(K tensor E^vee) = Hom(E, K). Stacks Project, Cohomology of Sheaves, Lemma lemma-dual-perfect-complex (tag 0A8C). Supplies the identification appearing as the left vertical arrow of the square.

Source reference. Original reference 0FPC: sources/stacks-chow/tags.txt, source lines 19597–19597. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08DQ: sources/stacks-chow/tags.txt, source lines 10645–10645. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0FP2: sources/stacks-chow/tags.txt, source lines 19587–19587. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original equation 08I2: sources/stacks-chow/tags.txt, source lines 10800–10800. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0A8V: sources/stacks-chow/tags.txt, source lines 12928–12928. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let K and E be objects of $D(\mathcal{O}_X)$ with E perfect. The diagram*

$$\begin{array}{ccc} H^0(X, K \otimes^{\mathbf{L}}_{\mathcal{O}_X} E^\vee) \times H^0(X, E) & \longrightarrow & H^0(X, K \otimes^{\mathbf{L}}_{\mathcal{O}_X} E^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} E) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_X(E, K) \times H^0(X, E) & \longrightarrow & H^0(X, K) \end{array}$$

commutes where the top horizontal arrow is the cup product, the right vertical arrow uses $\epsilon : E^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} E \to \mathcal{O}_X$ (Example 0FPC), the left vertical arrow uses Lemma 08DQ, and the bottom horizontal arrow is the obvious one.

Proof. We will abbreviate $\otimes = \otimes^{\mathbf{L}}_{\mathcal{O}_X}$ and $\mathcal{O} = \mathcal{O}_X$. We will identify E and K with $R\mathcal{H}om(\mathcal{O}, E)$ and $R\mathcal{H}om(\mathcal{O}, K)$ and we will identify $E^\vee$ with $R\mathcal{H}om(E, \mathcal{O})$.

Let $\xi \in H^0(X, K \otimes E^\vee)$ and $\eta \in H^0(X, E)$. Denote $\tilde{\xi} : \mathcal{O} \to K \otimes E^\vee$ and $\tilde{\eta} : \mathcal{O} \to E$ the corresponding maps in $D(\mathcal{O})$. By Lemma 0FP2 the cup product $\xi \cup \eta$ corresponds to $\tilde{\xi} \otimes \tilde{\eta} : \mathcal{O} \to K \otimes E^\vee \otimes E$.

We claim the map $\xi' : E \to K$ corresponding to ξ by Lemma 08DQ is the composition

$$E = \mathcal{O} \otimes E \xrightarrow{\tilde{\xi} \otimes 1_E} K \otimes E^\vee \otimes E \xrightarrow{1_K \otimes \epsilon} K$$

The construction in Lemma 08DQ uses the evaluation map (08I2) which in turn is constructed using the identification of E with $R\mathcal{H}om(\mathcal{O}, E)$ and the composition $\underline{\circ}$ constructed in Lemma 0A8V. Hence ξ' is the composition

$$\begin{aligned} E = \mathcal{O} \otimes R\mathcal{H}om(\mathcal{O}, E) &\xrightarrow{\tilde{\xi} \otimes 1} R\mathcal{H}om(\mathcal{O}, K) \otimes R\mathcal{H}om(E, \mathcal{O}) \otimes R\mathcal{H}om(\mathcal{O}, E) \\ &\xrightarrow{\underline{\circ} \otimes 1} R\mathcal{H}om(E, K) \otimes R\mathcal{H}om(\mathcal{O}, E) \\ &\xrightarrow{\underline{\circ}} R\mathcal{H}om(\mathcal{O}, K) = K \end{aligned}$$

The claim follows immediately from this and the fact that the composition $\underline{\circ}$ constructed in Lemma 0A8V is associative (insert future reference here) and the fact that ϵ is defined as the composition $\underline{\circ} : E^\vee \otimes E \to \mathcal{O}$ in Example 0FPC.

Using the results from the previous two paragraphs, we find the statement of the lemma is that $(1_K \otimes \epsilon) \circ (\tilde{\xi} \otimes \tilde{\eta})$ is equal to $(1_K \otimes \epsilon) \circ (\tilde{\xi} \otimes 1_E) \circ (1_{\mathcal{O}} \otimes \tilde{\eta})$ which is immediate. □